EVRYN Overarching Three-Phase Research Scope & Monograph Demarcation

PHASE 1

DELIVERED (Phase 1.14)

Multi-Page Causal Extraction

- Full-text PMC-OA JATS XML & PDF parsing with geometry
- Section discourse tagging & coordinate tracking
- Syntactic & agent-nominal causal trigger parsing
- Dual-sentence context resolution [Sc, Se]
- Bounded transitive multi-hop reasoning (<= 2 hops)
- Calibrated confidence verification & Platt scaling
- 100% provenance contract (doc, page, offsets, quotes)
- 20-paper authentic PMC benchmark validation

Grounded Graphs

PHASE 2

IN PROGRESS

Corpus Transfer & Adaptation

- Multi-document graph fusion across literature
- Cross-paper entity normalization & alias resolution
- 25-year longitudinal corpus trajectory tracking
- Causal mechanism drift & belief revision detection
- Continual SCM adaptation with zero forgetting
- Conformal semantic gating via PubMedBERT
- Empirical evaluation on temporal SCM streams
- Cross-study consensus & refutation modeling

Longitudinal Corpus

PHASE 3

PLANNED

Prospective Discovery Prediction

- Dynamic causal graph manifold representation
- Prospective emergent link forecasting
- Pre-publication hypothesis generation & ranking
- Multi-hop cross-domain mechanism completion
- Prospective validation on post-cutoff publications
- Biomedical discovery prioritization pipeline
- Comprehensive M.Tech defense monograph
- Reproducible open-source software release